

Explainable AI for Human Resource Analytics using Kolmogorov–Arnold Networks

Nishanth Reddy Dereddi, Pradyumna Reddy Dereddi, Sireesha Rodda*
*School of Computer Science and Engineering, GITAM Deemed to be University,
India.*

**corresponding author: srodde@gitam.edu*

Abstract. *One of the crucial responsibilities of human resource department is effectively predicting employee attrition. Over time, machine learning techniques are being successfully employed to identify patterns within data. HR analytics is an emerging area where machine learning is employed to address challenges faced in human resource management. Class imbalance problem and explainability still remain as challenges of these models. This paper investigates a real-world HR employee attrition dataset. When the proposed approach, based on Kolmogorov–Arnold Networks, is compared with a Multilayer Perceptron, it is observed that KAN provides inherent interpretability through feature-additive modeling.*

Keywords: *Explainable AI (XAI), Neural Networks, Imbalanced Learning, Kolmogorov-Arnold Networks*

1 Introduction

Human Resource (HR) Management manages the talent pool of employees and their lifecycle within an organization to boost productivity, enhance organizational knowledge, and optimize financial costs across recruitment and employee training [1]. Proactively identifying employees who might leave the organization may create space for organizations' employee retention strategies, thereby reducing losses and enhancing sustainability [2]. As a result, employee attrition prediction became a crucial factor in HR analytics [3].

The increasing availability of employee historical data has led to the development of Data-driven approaches for handling HR challenges [4]. Machine Learning has been effectively used to provide insights from historical employee data. Usually, employee data can be very rich and diverse, including personal, demographic, behavioral, and organizational data [5]. Given the complexity of the data, deep learning approaches can be better employed to uncover complex, nonlinear relationships in employee data [6]. Deep learning methods try to identify the complex relationships among attributes by fine-tuning neural networks.

The use of machine learning approaches has proven highly suitable for HR analytics, but employee data still poses some challenges. The attrition rate in the data is quite low [7]. Machine learning approaches tend to maximize the overall

accuracy of the predictive model. In the process, the majority class is favored over the minority class in employee attrition, thereby failing to identify employees at risk of Attrition [8] properly. Also, deep learning models suffer from their black-box nature, failing to justify their predictions. This occurrence undermines the organization's trust and transparency. Although post-hoc methods like SHAP and LIME are available to address explainability, they are approximate and do not provide sufficient help.

This paper attempts to work with a real-time HR Employee Attrition Dataset to address the challenges of learning in the presence of class imbalance and of attributing interpretability to neural networks. Kolmogorov-Arnold Networks (KAN) is compared with the conventional Multilayer Perceptron (MLP) to evaluate its predictive performance in light of class imbalance and interpretability. The experimental results on evaluation metrics measuring p classification performance, model interpretability, and feature importance stability. The results affirm that KAN provides a more reliable and explainable architecture for data-driven decision-making in human resource analytics.

2 Related Work

2.1 Employee Attrition Prediction in HR Analytics

The study of the factors leading to an employee's exit from a company is termed Employee Attrition Prediction. Early works relied on statistical techniques such as linear regression and survival analysis. These techniques provided interpretability for the models developed, but had limited capacity to capture the nonlinear relationships among employee attributes and their chances of voluntary exit from the organization.

As the volume of organizational data on employees grew, machine learning techniques became popular for predicting employee turnover. They improved accuracy, and the ensemble classifiers offered greater robustness across diverse datasets. The transparency, however, was limited, which limited their capacity for decision-making in HR applications.

2.2 Deep Learning for HR Analytics

Over the past few years, deep learning models have been extensively explored, mainly because of their ability to model employee attributes and the likelihood of their leaving the organization. Despite their strong predictive capability, employee databases are inherently imbalanced, with attrition cases accounting for only a small proportion of the overall dataset. Standard training of deep learning approaches may achieve high overall accuracy, but the minority nature of employee attrition may lead to a low recall rate. In addition, deep learning is often challenged by its black-box nature [9]. Approaches based on neural networks provide the required prediction but the interpretability is limited, thereby struggles to gain acceptance among the HR community.

2.3 Explainable AI in HR Analytics

To address transparency needs, explainable AI techniques have been incorporated into HR analytics. Retrospective techniques such as SHAP and LIME were used to provide interpretability to machine learning results [10]. But the approximate nature of these models made them sensitive to data perturbations, model instability, and other issues. XAI models, which are inherently interpretable, were sought.

Kolmogorov-Arnold Networks are recent additions to the deep learning family [11]. KANs replace conventional learnable weights with learnable nonlinear functions. The additive feature enables the KANs to inspect the feature contributions to learning tasks, including model stability and decent performance under class imbalance. Although KANs have been used in scientific tasks before, they remain largely underexplored in HR analytics.

3 Methodology

Overall system design, consisting of MLP and KAN models used predicting the outcome of the target variable, attrition status, is presented in this section. As both methods are based on Neural Networks, they will be compared for performance as well as interpretability in the presence of class imbalance.

3.1 Terminology

Consider the dataset $D = \{x_i, y_i\}_{i=1 \dots N}$. where $x_i \in R^d, y \in \{0,1\}$ denotes the feature vector corresponding to the i-th employee, consisting of d demographic, organizational, and behavioral attributes, and $y_i \in \{0, 1\}$ represents the class label. Here, $y_i = 1$ indicates employee attrition, while $y_i = 0$ denotes employee retention. The dataset exhibits class imbalance, with attrition cases forming the minority class.

The objective is to learn a function $f: R_d \rightarrow [0, 1]$ that estimates the probability of employee attrition.

A Multilayer Perceptron is a feed-forward neural network composed of several fully connected layers. For an MLP with L hidden layers, the forward propagation is defined as:

$$h(l) = \sigma(w^{(l)} h^{(l-1)} + b^{(l)}), l = 1, \dots, L - 1. \quad (1)$$

$$\hat{y} = w^{(l)} h^{(l-1)} + b^{(l)} \quad (2)$$

Although MLPs are effective in modeling nonlinear relationships, it is not possible to interpret the feature contributions. Kolmogorov–Arnold Networks (KANs) are based on the Kolmogorov–Arnold representation theorem, which states that any continuous multivariate function can be represented as a finite

sum of compositions of univariate functions. According to KAN, the target function is modeled as:

$$f(x) = \sum_{j=1}^d g_j(x_j) \quad (3)$$

where x_j denotes the j -th input feature and $g_j(\cdot)$ is a learnable univariate nonlinear function associated with that feature.

Each $g_j(\cdot)$ is parameterized using spline-based or kernel-based functions, allowing flexible nonlinear modeling while preserving explicit feature-wise decomposition. The predicted probability of Attrition is given by:

$$\hat{y} = \sigma(\sum_{j=1}^d g_j(x_j) + b) \quad (4)$$

where b is a bias term and $\sigma(\cdot)$ corresponds to the sigmoid activation function.

Both MLP and KAN use on the same binary cross-entropy loss to train the model. However, the additive structure of KAN enables direct attribution of predictions to individual input features.

3.2 Handling Class Imbalance

A class-weighted binary cross-entropy loss is employed for both models, to address the problem of class imbalance.

$$L_{WBCE} = -\frac{1}{N} \sum_{i=1}^N w_1 \hat{y}_i \log(\hat{y}_i) + w_0 (1 - \hat{y}_i) \log(1 - \hat{y}_i) \quad (5)$$

where w_1 and w_0 are weights inversely proportional to the frequency of the minority and majority classes, respectively. This loss function places greater weight on false negatives and improves recall for attrition cases.

3.3 Interpretability and Feature Importance

Interpretability is hard to provide for MLPs, as their feature importance is distributed across layers. On the other hand, KAN inherently provides interpretability. After the univariate functions $g_j(x_j)$ are trained, they represent how each feature contributes to the outcome. This can be used to provide visualization, feature importance estimation, and better performance than post-hoc explanation techniques.

4 Experimental Results and Analysis

4.1 Dataset Description

The empirical results are obtained from on a real-world Human Resource Analytics dataset consisting of employee-level records spanning demographic,

professional, and organizational attributes. The dataset includes features such as Age, Gender, Department, Job Role, Monthly Income, Job Level, Years at Company, Years Since Last Promotion, Overtime, Work–Life Balance, Job Satisfaction, and Performance Rating. The target variable, Attrition, is binary, indicating whether the employee left or stayed with the organization.

The dataset under consideration has clear imbalance in class representations. Approximately 16.1% of the records correspond to employee attrition, while 83.9% represent non-attrition cases. This distribution is similar to real-world situations in HR management. The class imbalance problem poses challenges for predictive modeling, especially in minimizing the false negative ratio.

4.2 Experimental Setup Overview

The Employee Attrition dataset is suitably preprocessed for handling categorical and continuous variables. Categorical variables are subjected to one-hot encoding whereas continuous variables are normalized. The data is then divided into training and validation datasets. Both models are trained under identical conditions: same training and validation datasets, same optimization strategies as well as class weights, to enable fair comparison of performance.

Model performance is evaluated using Accuracy, Precision, Recall, F1-score, Specificity, False Negative Rate (FNR), and the Area Under the ROC Curve (AUC). ROC and Precision–Recall curves are also examined to provide a complete evaluation of classification performance.

4.3 Classification Performance Comparison

The comparison of quantitative performance between MLP and KAN is summarized in Table 1. The MLP achieves higher overall accuracy and specificity, indicating strong performance on the majority non-attrition class. However, its recall for the attrition class is low (0.41), resulting in a high false negative rate (0.59). It can be inferred that a large proportion of employees who eventually leave are not identified in advance.

KAN significantly improves minority-class detection, achieving a recall of 0.76 and reducing the false negative rate to 0.24. Although the overall accuracy of KAN is marginally lower, its superior F1-score and AUC indicate a better balance between sensitivity and precision under class imbalance.

Table 1. Performance Comparison of MLP and KAN

Metric	MLP	KAN
Accuracy	0.82	0.79
Precision (Attrition)	0.55	0.71
Recall (Attrition)	0.41	0.76
F1-score	0.47	0.73

Specificity	0.90	0.83
False Negative Rate (FNR)	0.59	0.24
AUC	0.68	0.81

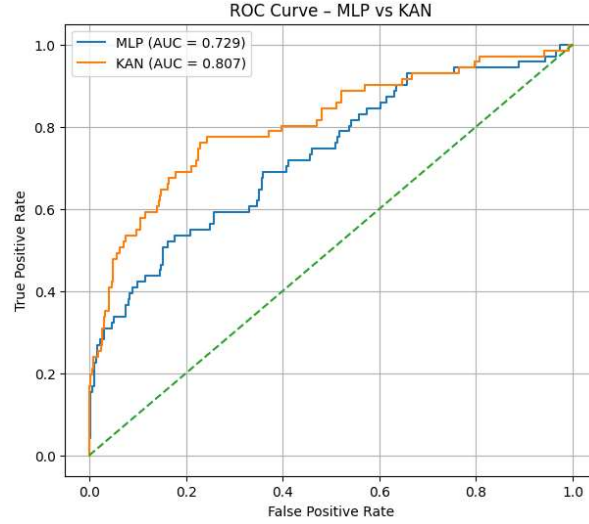


Fig. 1. ROC Curves for MLP and KAN

4.4 ROC and Precision–Recall Analysis

The ROC curves obtained from the results of both KAN and MLP are shown in Figure 1. It can be seen that KAN returns a higher mean AUC than MLP, of 0.81 as compared to 0.68 respectively. The bootstrapped confidence intervals further show the robustness of KAN, with tighter bounds and reduced variability.

The Precision-Recall curves, as presented in Figure 2, show how KAN keeps maintaining higher precision over a broader range of recall values. This indicates a more reliable performance of KAN across varying decision thresholds. This is particularly important in HR analytics, where intervention capacity may vary across organizations.

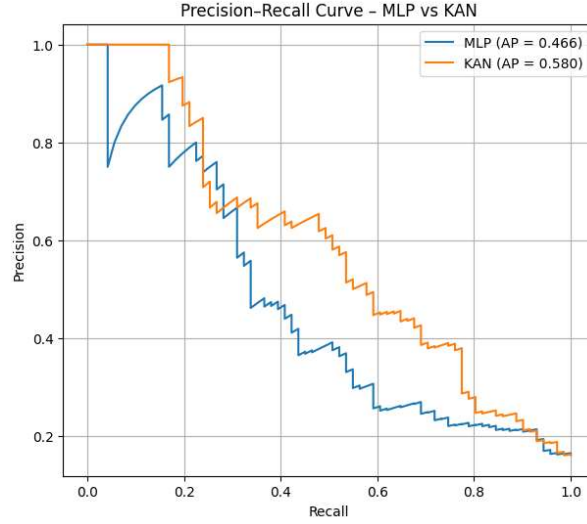


Fig. 2. Precision-Recall Curves for MLP and KAN

4.5 Impact of Class Imbalance

The experimental results confirm that KAN is more resilient to class imbalance than the MLP. MLP returns better accuracy but does not perform better when it comes to performance metrics like AUC and F1-score which are more impervious to the class imbalance problem. On the other hand, KAN effectively captures minority-class patterns. The pattern can be observed from the higher scores of AUC, recall, F1-score and lower false negative rate. The empirical results indicate the KAN is more suitable for applications suffering from class imbalance problem than MLP.

Table 2. Top Features by KAN Importance Scores

Feature	KAN Score
OverTime	0.723203
MonthlyIncome	0.624771
YearsSinceLastPromotion	0.540920
NumCompaniesWorked	0.453885
TotalWorkingYears	0.446830
Department	0.446248
YearsWithCurrManager	0.415704
EnvironmentSatisfaction	0.384605
JobSatisfaction	0.384289

4.6 Interpretability and Feature Importance Analysis

Table 2 presents the feature-wise functions learned by KAN. The importance of every feature and its effect on employee attrition is provided intrinsically as KAN score. It can be observed that features such as OverTime, Monthly Income, Years Since Last Promotion, Job Level, and Work–Life Balance are key features contributing to employee attrition. These patterns are stable across runs and align closely with the established HR process.

4.7 Analysis

From the HR management viewpoint, it is more important to identify false negatives for preventing employee attrition. The at-risk employees can be proactively identified, as implied by the improved recall rate. In order to remedy the employee attrition, they may be offered targeted incentives such as reduced workload, compensations, and workload management. The interpretability feature enables the policy-makers to draft a more transparent and data-driven HR policy.

From the results, it can be observed that KAN provides better results than MLP across recall, F1-measure and AUC parameters. The robustness of the KAN approach can be seen in how it reduces the false negatives while providing a stable and trustworthy approach for employee attrition prediction.

5 Conclusions and Future Directions

The problem of Employee Attrition Prediction is addressed using Kolmogorov–Arnold Networks in this paper and compared with a conventional Multilayer Perceptron (MLP). Experimental results indicate that KAN achieved better performance metrics, especially when the data was imbalanced. In such cases, it was observed that KAN achieved higher AUC, ROC, and recall, and lower false-negative rates than MLP. These can be attributed to KAN's suitability for identifying employee attrition in the organization and for taking preventive measures therein. Because KANs use feature-additive modeling, it is possible to quantify the contributions of individual features. The results indicate that key attrition drivers, such as overtime, compensation, promotion history, and work–life balance, drive the employee attrition rate upwards. Overall, KAN offers a reliable and explainable solution for HR analytics.

The work may be extended to a more diverse set of employees over longer durations to improve the generalization ability of KANs. Future work could consider longitudinal employee data and larger datasets to improve generalization. The framework can be extended to medical diagnostics, where class imbalance is common. But establishing clinical trust is paramount to acceptance of clinical decision support systems.

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